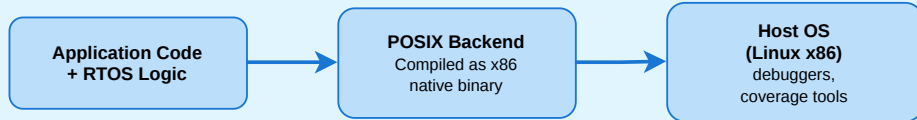


Layer 1: Host-Native Execution

native_sim (Zephyr) / POSIX simulator port (FreeRTOS)



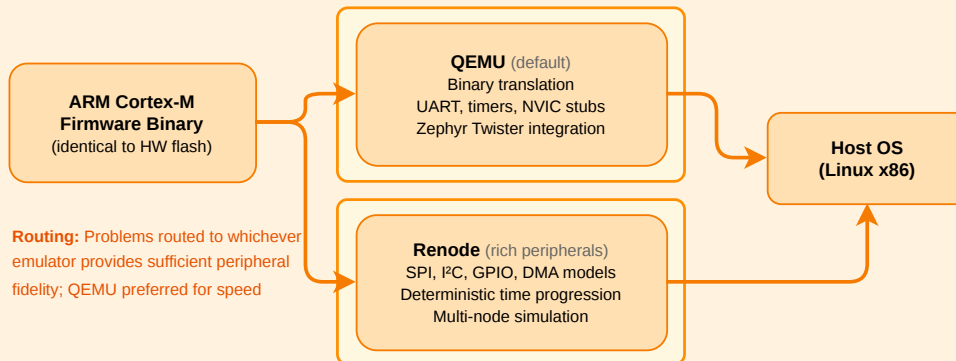
- ✗ No ISA fidelity
- ✗ No peripheral modeling
- ✓ Full host tooling

Problem Types:

- Task scheduling bugs
- Synchronization primitives
- Data structure correctness
- Pure RTOS logic errors

Layer 2: System-Level Emulation

Instruction-level emulation of target ISA + software peripheral models



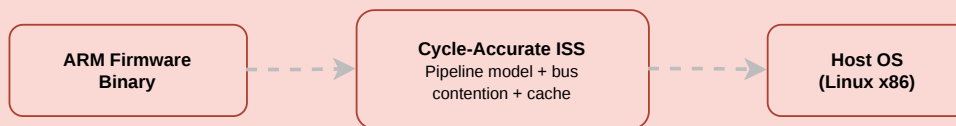
- ✓ Real ARM ISA execution
- ✓ Interrupt controller model
- ✓ Peripheral SW models
- ✗ No cycle-accurate timing

Problem Types:

- ISA-dependent behavior
- Interrupt handling bugs
- Memory-mapped I/O issues
- Peripheral driver logic
- UART/SPI/DMA config errors

Layer 3: Cycle-Accurate Simulation — Future Work

Pipeline model, bus contention, memory access latencies at clock-cycle granularity



- ✓ Exact instruction timing
- ✓ Pipeline behavior
- ✗ Limited open-source tooling

Problem Types (future):

- Pipeline-dependent races
- Timing side-channel vulns
- Exec-time-dependent control flow

Hardware Fidelity

Execution Speed